Lecture 4

Covering spaces and rotations

Objectives. *We generalise the circle projection of Lecture 3 to covering maps. Their local inverse branches lift paths and homotopies. For a simply connected covering space, lifted endpoints determine loop classes. We apply this to rotations: one full turn is nontrivial, and two full turns contract.*

By the end of this lecture you will be able to:

- Prove path and path-homotopy lifting for a covering map.
- Recover loop classes from a simply connected covering space.
- Compute $\pi_1(\mathbb{R}\mathrm{P}^n)$ for $n \geq 2$ and $\pi_1(\mathrm{SO}(3))$.
- Distinguish the lifted paths of a 2π and a 4π rotation.

Unit 4A Coverings, lifted endpoints and rotations

4.1 Coverings and local inverse branches

Definition 4.1 (Covering map)**.** A *covering map* is a continuous surjection $p\colon \widetilde{X} \to X$ such that each $x \in X$ has an open neighbourhood U with

$$p^{-1}(U) = \coprod_{\alpha} V_\alpha, \qquad p|_{V_\alpha}\colon V_\alpha \longrightarrow U \text{ a homeomorphism.}$$

The symbol $\coprod$ denotes a disjoint union. The sets V_α are open in $\widetilde{X}$. We call U *evenly covered* and each V_α a *sheet over* U.

The fibre $p^{-1}(\{x\})$ is the inverse image of the point, as in Lecture 2. It contains one point in each sheet over U. In particular, the fibre has the discrete subspace topology: every one of its points is open in the fibre. Its cardinality is constant as x varies in an evenly covered set. The figure shows the local structure.

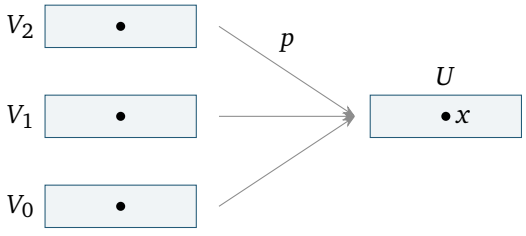


Figure 4.1: Each sheet maps homeomorphically onto the same open set U. The marked points belong to one fibre.

For example, the projection $u \mapsto e^{2\pi i u}$ from $\mathbb{R}$ to S^1 is a covering. The semicircles U_r of §3.2 have inverse images $\coprod_{k\in\mathbb{Z}}(J_r + k)$. Each restriction has inverse $b_r + k$. The map $z \mapsto z^m$ on S^1, for an integer $m \geq 1$, is an m-sheeted covering. An open arc of angle less than 2π has m disjoint inverse arcs, obtained by dividing its angles by m and adding $2\pi k/m$ for $k = 0, \ldots, m-1$. Lecture 3 chose a continuous phase along a circle path, rather than a single inverse branch on the whole circle. A loop of winding one raises that phase by one turn. In the helix model, the real coordinate is the height, and projection forgets the height. For the double cover, following a loop of winding one exchanges the two fibre points. A second traversal returns to the initial point upstairs. The figure compares these global behaviours.

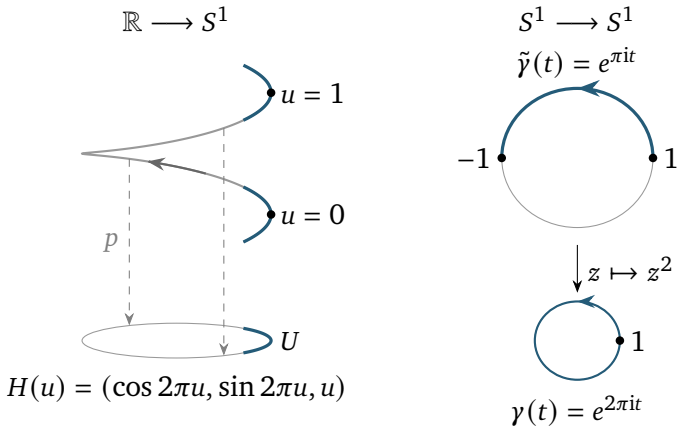


Figure 4.2: The helix projects vertically to the circle and rises one unit per positive turn. For $z \mapsto z^2$, a positive semicircle upstairs projects to a full circle downstairs and ends at the other point of the fibre.

Definition 4.2 (Local homeomorphism)**.** A *local homeomorphism* restricts near each domain point to a homeomorphism onto an open subset of the target.

Every covering has this property. The converse fails, even for a surjection. Consider

$$q\colon (0, 3\pi) \longrightarrow S^1, \qquad q(\theta) = e^{i\theta}.$$

Small intervals give local inverses. The fibre over 1 is $\{2\pi\}$, but the fibre over $e^{i\varepsilon}$ has two points whenever $0 < \varepsilon < \pi$. Every neighbourhood of 1 contains such points. Thus 1 has no evenly covered neighbourhood. The figure marks the missing endpoint of the extra inverse branch.



Figure 4.3: The truncated angle interval contains one preimage of 1, but two of $e^{i\varepsilon}$. Open endpoints prevent a second branch at 1.

The missing endpoint also prevents a path lift from existing on the whole interval. For $\gamma(t) = e^{-2\pi i t}$, the only possible initial value is 2π. Any lift would differ from $2\pi(1-t)$ by a continuous integer multiple of 2π. This integer is zero at the start, so it is zero everywhere. The forced endpoint is 0, outside the domain of q. Thus local inverse branches alone do not guarantee path lifting.

4.2 Lifting paths and their homotopies

Definition 4.3 (Lift). A *lift* of $f\colon Y \to X$ is a continuous map $\tilde{f}\colon Y \to \widetilde{X}$ with $p \circ \tilde{f} = f$.

Write $I = [0, 1]$.

Theorem 4.4 (Path and path-homotopy lifting). *For a covering p, a path $\gamma\colon I \to X$ has a unique lift after specifying $\tilde{\gamma}(0)$ above $\gamma(0)$. A homotopy $h\colon I^2 \to X$ has a unique lift after a lift of its bottom edge has been specified. If h fixes its path endpoints, so does its lift.*

Proof. First, uniqueness of a path lift follows locally. On a subinterval mapping into an evenly covered U, any lift stays in one sheet: the interval is connected and the sheets are disjoint and open. The sheet is determined by one lifted value. Its inverse branch determines the whole subinterval.

For existence, pull back evenly covered sets to an open cover of I. The Lebesgue-number lemma of Lemma 3.5 gives a finite subdivision such that each closed subinterval maps into one such set. On the first subinterval use the sheet containing the prescribed initial point. On each successive subinterval use the sheet containing the preceding endpoint. The formulas agree at the joins and paste continuously. Local uniqueness on successive subintervals gives uniqueness on I.

For a homotopy, pull the same open cover back to the compact square. Choose a grid with each cell mapping into an evenly covered set. Extend the bottom lift row by row, from left to right. Choose on each cell the inverse branch whose sheet contains the already assigned lower-left value. On its bottom and left edges, path-lift uniqueness forces agreement with every previously assigned value. There are no other previously assigned points on the cell. Finite pasting gives a continuous lift on the square. The same cell order proves uniqueness. This is the grid construction of Lemma 3.6, with sheets in place of integer translates.

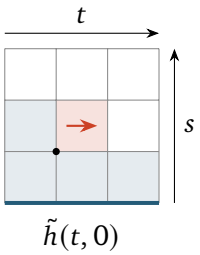


Figure 4.4: The new inverse branch is chosen at the marked corner. Uniqueness on shared edges makes it agree with the shaded region.

Finally, if an edge of h is constant, its lift is a path in a discrete fibre. A continuous image of the connected interval is connected, so this path is constant. Apply this to the two endpoint edges. ∎

In particular, endpoint-fixed homotopic paths have lifts with the same endpoint when their initial lifts agree. The induced homomorphism p_* is injective: lift a based contraction of a projected loop. The lift is a based contraction of the original loop. Also,

$$[\gamma] \in p_*\pi_1(\widetilde{X}, \tilde{x}_0) \quad \Longleftrightarrow \quad \text{the lift of } \gamma \text{ from } \tilde{x}_0 \text{ closes.}$$

For the reverse implication project the closed lift. For the forward implication compare γ with the projection of a representative loop upstairs and lift their based homotopy.

4.3 Fibres, loop classes and deck transformations

For the circle projection, translations $u \mapsto u + k$ with $k \in \mathbb{Z}$ preserve the projected point. Composing translations adds their integers, just as concatenating loops adds winding numbers. We use transformations preserving a covering projection to recover the multiplication of loop classes from lifted endpoints.

Definition 4.5 (Deck transformation). A *deck transformation* of p is a homeomorphism $D\colon \widetilde{X} \to \widetilde{X}$ with $p \circ D = p$. Their group under composition is written $\mathrm{Deck}(p)$.

On a path-connected covering space, a deck transformation is determined by its value at one point. Indeed, for a path a from that point to any other point, $D \circ a$ is the unique lift of $p \circ a$ from the specified image point. In particular, a deck transformation fixing one point is the identity. The deck group acts *transitively* on a fibre if for any two points of that fibre some deck transformation sends the first to the second.

Corollary 4.6 (Counting a simply connected covering). *Let $p\colon \widetilde{X} \to X$ be a covering with $\widetilde{X}$ simply connected. Fix $\tilde{x}_0$ above x_0. Sending a loop class to its lifted endpoint gives a bijection*

$$\pi_1(X, x_0) \longrightarrow p^{-1}(\{x_0\}).$$

If the deck group acts transitively on this fibre, it is isomorphic to $\pi_1(X, x_0)$.

Proof. The endpoint depends only on the based class by homotopy lifting. Every fibre point can be joined to $\tilde{x}_0$, since simple connectedness includes path-connectedness. Projecting this path proves surjectivity. If two lifted paths have the same endpoint, join the first to the reverse of the second. This loop contracts in $\widetilde{X}$. Its projection shows that the two original loop classes agree. This proves injectivity.

Under transitivity, let D_γ be the unique deck transformation taking $\tilde{x}_0$ to the endpoint of the lift $\tilde{\gamma}$. The preceding bijection and deck uniqueness give a bijection of groups. The path $D_\gamma \circ \tilde{\delta}$ projects to δ, because $p \circ D_\gamma = p$. It starts at the endpoint of $\tilde{\gamma}$. Path-lift uniqueness therefore makes it the continuation of the lift of $\gamma \cdot \delta$ after $\tilde{\gamma}$. The concatenated lift ends at

$$D_\gamma(D_\delta(\tilde{x}_0)), \qquad D_{\gamma\cdot\delta} = D_\gamma \circ D_\delta.$$

Thus the bijection preserves multiplication with the course's first-path-first convention. The isomorphism uses the chosen lifted base point. ∎

The figure shows this continuation for circle loops of windings 2 and -1.

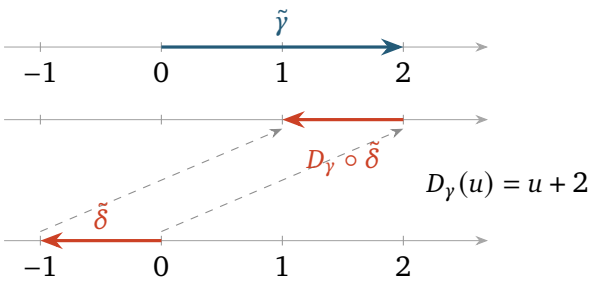


Figure 4.5: The lift of the second loop starts at 0 and ends at -1. Translation by 2 moves it to a path from 2 to 1, which continues the first lift in the order γ then δ.

For example, the deck transformations of $\mathbb{R} \to S^1$ are exactly $u \mapsto u + k$ for $k \in \mathbb{Z}$. These are deck transformations, act transitively on the fibre, and exhaust the possibilities by uniqueness from the image of 0. They recover $\pi_1(S^1, 1) \cong \mathbb{Z}$. The deck group $\mathbb{Z}$ is the first example of two notions used from Lecture 5 onward.

Definition 4.7 (Subgroup)**.** A *subgroup* of a group G is a subset $H \subseteq G$ containing the identity and closed under multiplication and inverses.

Definition 4.8 (Abelian group)**.** A group G is *abelian* if $gh = hg$ for every $g, h \in G$.

For example, $m\mathbb{Z}$ is a subgroup of the additive group $\mathbb{Z}$ for every integer $m \geq 0$. The group $\mathbb{Z}$ is abelian because addition commutes.

4.4 The antipodal cover and rotations

Proposition 4.9 (Spheres of dimension at least two · *sketched*)**.** *The sphere S^n is simply connected for $n \geq 2$.*

Proof sketch. Short spherical arcs join nonantipodal points. A path via a third point joins antipodes, so the sphere is path-connected.

A loop missing a point contracts after stereographic projection to $\mathbb{R}^n$. An arbitrary continuous loop need not miss a point. We first replace the loop by a homotopic spherical polygon, whose image lies in finitely many proper linear subspaces.

First, cover the loop by open hemispheres. Compactness and Lemma 3.5 give a finite subdivision whose pieces each lie in one hemisphere. Replace each piece by the shortest spherical arc joining its endpoints in that hemisphere. Normalizing their straight interpolation gives the homotopy: its scalar product with the hemisphere centre stays positive.

Next, the resulting polygon lies in finitely many two-dimensional linear subspaces. For $n \geq 2$ these are proper subspaces of $\mathbb{R}^{n+1}$ and their union misses a point of the sphere. Finally, stereographic projection from that point identifies its complement with $\mathbb{R}^n$, where the polygon contracts with its base point fixed. The formulas of Lecture 1 apply in n coordinates. This sketches the sphere result of Hatcher, Proposition 1.14, using polygonal approximation.

This also distinguishes the plane from higher-dimensional Euclidean spaces, given Proposition 4.9 at its sketched level. For every $n \geq 2$, the radial homotopy is given by

$$h\colon (\mathbb{R}^n \setminus \{0\}) \times I \longrightarrow \mathbb{R}^n \setminus \{0\}, \qquad h(x,t) = \left(1 - t + \frac{t}{|x|}\right) x.$$

Its positive scalar factor keeps the image away from the origin. It fixes the unit sphere. The final map's value at x is $x/|x|$. Thus $\mathbb{R}^n \setminus \{0\}$ has the fundamental group of S^{n-1}, by Lecture 3's homotopy-invariance argument. For $n \geq 3$ this group is trivial, whereas the punctured plane has group $\mathbb{Z}$ by Lecture 3. A homeomorphism from $\mathbb{R}^2$ to $\mathbb{R}^n$ would restrict to a homeomorphism after deleting 0 and its image. Translation identifies the latter puncture with 0, contradicting these different groups. Hence $\mathbb{R}^2$ is not homeomorphic to $\mathbb{R}^n$ for $n \geq 3$.

Recall from Lecture 2 that $\mathbb{R}\mathrm{P}^n$ identifies antipodal points of S^n. The quotient map is a covering. To see this, for $x \in S^n$ take the cap $V = \{u \in S^n : u \cdot x > 1/2\}$. The inverse image of its quotient image is $V \coprod (-V)$. The two caps are disjoint. Write p for this quotient map. For every open $W \subseteq S^n$,

$$p^{-1}(p(W)) = W \cup (-W).$$

This set is open, so $p(W)$ is open by the quotient topology. A map taking open sets to open sets is called an *open map*. Each cap therefore maps homeomorphically onto the same open quotient set. These sets cover $\mathbb{R}\mathrm{P}^n$. The figure shows the two caps over one neighbourhood.

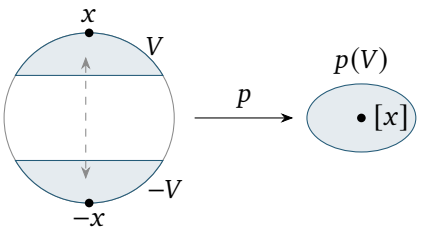


Figure 4.6: The two disjoint caps V and $-V$ lie above the same quotient neighbourhood. The antipodal map exchanges the two sheets.

Write $\mathbb{Z}_2$ for the integers modulo two.

Theorem 4.10 (Projective space and rotations · *proved given S^n simply connected*)**.** *For $n \geq 2$, the antipodal cover gives $\pi_1(\mathbb{R}\mathrm{P}^n) \cong \mathbb{Z}_2$. Consequently $\pi_1(\mathrm{SO}(3)) \cong \mathbb{Z}_2$.*

Proof. The sphere is simply connected by Proposition 4.9, at the sketched level just given. The identity and antipodal map act transitively on the two-point fibre. Deck uniqueness excludes any other transformation. The antipodal map has square the identity, so Corollary 4.6 gives $\mathbb{Z}_2$. The homeomorphism $\mathbb{R}\mathrm{P}^3 \cong \mathrm{SO}(3)$ was proved in Proposition 2.22 by the axis–angle construction. ∎

The nontrivial class has a representative with a direct geometric meaning. Take $x = e_1$ and the path

$$\tilde{\gamma}(t) = \cos(\pi t)e_1 + \sin(\pi t)e_2 \qquad (t \in I)$$

on S^n. Its endpoint is $-x$, so its projection to $\mathbb{R}\mathrm{P}^n$ is a loop whose lift does not close. A point of projective space is a line through the origin. This projected loop turns the line through a half-turn. Traversing it twice lifts to a full great circle, which closes and contracts on S^n for $n \geq 2$. The figure compares the sphere path with its lines.

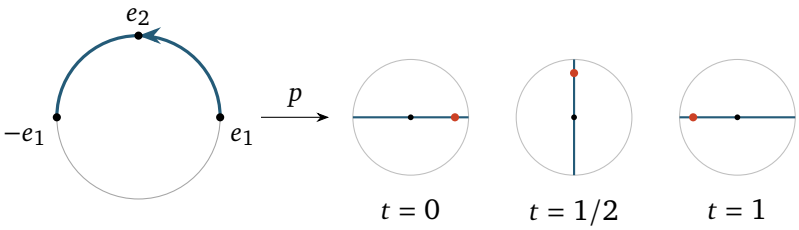


Figure 4.7: The upper semicircle joins e_1 to $-e_1$ on S^n. The lines at $t = 0$, $t = 1/2$ and $t = 1$ show the projected loop returning to its initial line after a half-turn.

The circle case is different: $\mathrm{SO}(2) \cong S^1$ has fundamental group $\mathbb{Z}$. The general result $\pi_1(\mathrm{SO}(m)) \cong \mathbb{Z}_2$ for $m \geq 3$ is cited from Nakahara, Table 4.1; only $m = 3$ is proved in this lecture.

4.5 The half-angle covering and Pauli matrices

The group computed in Theorem 4.10 has two elements. To determine the class of a particular rotation loop, we need an explicit covering formula and its lifted path. The half-angle coordinates associate a rotation through angle φ about a unit vector $n \in \mathbb{R}^3$ with

$$(a, v) = \left(\cos\frac{\varphi}{2}, \sin\frac{\varphi}{2}\, n\right) \in S^3.$$

Substituting $\varphi = 2\theta$ in Lecture 2's axis–angle formula gives the following rotation. Write a point of S^3 as (a, v), where $a \in \mathbb{R}$, $v \in \mathbb{R}^3$ and $a^2 + |v|^2 = 1$. The corresponding rotation is

$$w \longmapsto (a^2 - |v|^2)w + 2(v \cdot w)v + 2a(v \times w).$$

When $v = 0$, this is the identity. Otherwise, substituting $a = \cos\theta$ and $v = \sin\theta\, n$ gives Lecture 2's rotation $R(2\theta n)$ about the unit axis n. The formula is unchanged by $(a, v) \mapsto (-a, -v)$, so it induces a continuous map from $\mathbb{R}\mathrm{P}^3$ to $\mathrm{SO}(3)$.

To identify that map, choose representatives with $a \geq 0$ and $0 \leq \theta \leq \pi/2$. The corresponding axis–angle vector is

$$v \longmapsto \frac{2\arcsin|v|}{|v|} v \quad (v \neq 0), \qquad 0 \longmapsto 0.$$

Its radius increases strictly and continuously from 0 to π, so it is a bijection from the hemisphere to the closed angle ball. Continuity at $v = 0$ follows because the image norm is $2\arcsin|v| \to 0$. By Proposition 2.22, composing it with the angle-ball rotation identifies only antipodal boundary points. Thus the induced map $\mathbb{R}\mathrm{P}^3 \to \mathrm{SO}(3)$ is a continuous bijection and is a homeomorphism by Corollary 2.5. The displayed rotation formula is therefore the antipodal covering followed by this homeomorphism, and hence is itself a covering $S^3 \to \mathrm{SO}(3)$. Its radial coordinates differ from Lecture 2's linear hemisphere rescaling.

Definition 4.11 (Special unitary group)**.** The *special unitary group* $\mathrm{SU}(2)$ consists of complex unitary 2×2 matrices of determinant one.

Its matrices have the form below, identifying it with S^3:

$$U(a, v) = \begin{pmatrix} a - \mathrm{i}v_3 & -v_2 - \mathrm{i}v_1 \\ v_2 - \mathrm{i}v_1 & a + \mathrm{i}v_3 \end{pmatrix}, \qquad a^2 + |v|^2 = 1.$$

Indeed, orthonormal columns and determinant one give the form $\left(\begin{smallmatrix} z & w \\ -\bar{w} & \bar{z} \end{smallmatrix}\right)$ with $|z|^2 + |w|^2 = 1$. Taking real and imaginary parts gives the displayed coordinates and a continuous inverse.

We express the same matrix using the Pauli matrices, with the conventions of *Linear Algebra*:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_y = \begin{pmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix}, \qquad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

For $w = (w_1, w_2, w_3) \in \mathbb{R}^3$, write $w \cdot \sigma = w_1\sigma_x + w_2\sigma_y + w_3\sigma_z$. Matrix multiplication gives

$$(v \cdot \sigma)(w \cdot \sigma) = (v \cdot w)\mathbb{1} + \mathrm{i}(v \times w) \cdot \sigma.$$

In particular, $(n \cdot \sigma)^2 = \mathbb{1}$ for $|n| = 1$. In the half-angle coordinates, the even and odd powers in the exponential series therefore give

$$U(a, v) = a\mathbb{1} - \mathrm{i}v \cdot \sigma = \cos\frac{\varphi}{2}\mathbb{1} - \mathrm{i}\sin\frac{\varphi}{2}\, n \cdot \sigma = \exp\left(-\frac{\mathrm{i}\varphi}{2}\, n \cdot \sigma\right).$$

The same multiplication rule relates the matrix to its spatial rotation. Its triple-product consequence is

$$(v \cdot \sigma)(w \cdot \sigma)(v \cdot \sigma) = 2(v \cdot w)v \cdot \sigma - |v|^2 w \cdot \sigma.$$

Expanding the conjugation with this identity gives

$$\begin{aligned} U(w \cdot \sigma)U^\dagger &= \big((a^2 - |v|^2)w + 2(v \cdot w)v + 2a(v \times w)\big) \cdot \sigma \\ &= (Rw) \cdot \sigma. \end{aligned}$$

Here R is exactly the rotation displayed at the start of the section. Replacing U by $-U$ leaves the conjugation unchanged.

4.6 A full turn and a double turn

We compute the class of rotation through angle $2\pi t$ about the third axis, for $t \in I$.

The lifted path from the identity for the chosen rotation is

$$(a(t), v(t)) = (\cos \pi t,\ \sin \pi t\, e_3), \qquad U(t) = \begin{pmatrix} e^{-\pi i t} & 0 \\ 0 & e^{\pi i t} \end{pmatrix}.$$

At $t = 1$ it ends at $(-1, 0)$, or $-\mathbb{1}$ in matrix form. Its projection is closed, but its lift is not. By Corollary 4.6, this is the nontrivial loop class in $\mathrm{SO}(3)$.

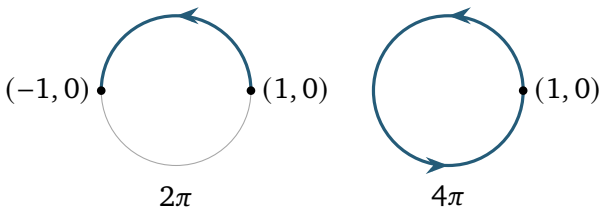


Figure 4.8: A full turn lifts from $(1, 0)$ to $(-1, 0)$ along a semicircle in S^3. A double turn completes the great circle and closes upstairs.

For a 4π rotation, replace t by $2t$ in the lifted formula. The lift ends at $(1, 0)$ and is a loop in the simply connected sphere. It contracts there, so its projection contracts in $\mathrm{SO}(3)$. The rotation ball of Lecture 2 represents the same full-turn loop by the piecewise representatives

$$b(t) = \begin{cases} 2\pi t\, e_3, & 0 \le t \le \frac{1}{2}, \\ (2\pi t - 2\pi)e_3, & \frac{1}{2} < t \le 1. \end{cases}$$

At the join, the two one-sided values are antipodal boundary representatives of the same rotation. Thus the path runs from the centre to πe_3, continues from $-\pi e_3$, and returns to the centre. A double turn traverses this loop twice. The spherical contraction can be written explicitly. In coordinates (a, v_1, v_2, v_3), put

$$q_s(t) = \Big(\cos \frac{\pi s}{2} \cos(2\pi t),\ 0,\ \sin \frac{\pi s}{2},\ \cos \frac{\pi s}{2} \sin(2\pi t)\Big) \quad (s, t \in I).$$

Every q_s is a loop on S^3. The family starts at the double-turn lift and ends at the constant point $(0, 0, 1, 0)$. Its base point moves. Lecture 3's change-of-base-point argument turns this free contraction into a based contraction, so its projection is the identity loop class. The covering proof above also supplies a based contraction directly by simple connectedness.

The figure shows the identified boundary points and the corresponding closed lift of the double turn.

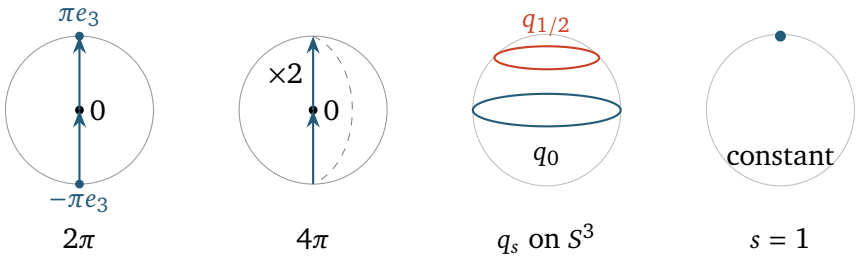


Figure 4.9: The ball section uses its antipodal boundary identification. A full turn passes through that identification once, while a double turn lifts to a full great circle that contracts on S^3.

A belt can be modelled by an oriented frame along its centre line. For a straight belt with k full twists, an explicit strip is

$$B_k(t,r) = \big(t, r\cos(2\pi kt), r\sin(2\pi kt)\big), \qquad t \in I, \quad -\tfrac{1}{8} \le r \le \tfrac{1}{8}.$$

At $r = 0$, its unit tangent is e_1. The width direction rotates through $2\pi kt$ about e_1. The tangent, width and their cross product form a path of oriented frames, hence a path in $\mathrm{SO}(3)$. For integer k the end frames agree, so it is a loop. A continuous motion of such frames fixing their end orientations gives an endpoint-fixed homotopy of that loop. The figure compares one and two twists with their frame paths.

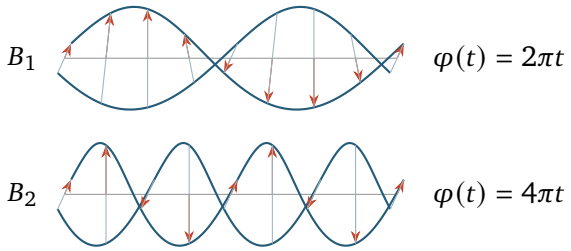


Figure 4.10: The strips are the exact models B_1 and B_2, with width directions marked along their centre lines. Their frame loops have respectively nonclosed and closed lifts to S^3.

For $k = 1$ the nontrivial loop class obstructs a deformation to constant frames. For $k = 2$ the frame loop contracts. This frame statement alone does not construct a motion of an embedded belt with fixed end positions.

Physical input. For a spin-1/2 degree of freedom, a pure state is represented by a nonzero vector $\psi \in \mathbb{C}^2$. Vectors differing by an overall phase represent the same isolated state. A rotation through φ about a unit axis n acts by

$$\psi \longmapsto \exp\left(-\frac{\mathrm{i}\varphi}{2}\, n\cdot\sigma\right)\psi.$$

The spin operators are $(\hbar/2)\sigma_x$, $(\hbar/2)\sigma_y$ and $(\hbar/2)\sigma_z$. Coherent alternatives combine by addition of their state amplitudes.

We can now derive the transformation of the spin expectation vector. Define its dimensionless components by

$$r_j = \frac{\psi^\dagger \sigma_j \psi}{\psi^\dagger \psi} \qquad (j = x, y, z).$$

For every $w \in \mathbb{R}^3$, the conjugation identity applied to $U^\dagger$ gives

$$w \cdot r' = \frac{\psi^\dagger U^\dagger (w \cdot \sigma) U \psi}{\psi^\dagger \psi} = (R^{-1} w) \cdot r = w \cdot (Rr).$$

Thus the expectation vector rotates by R. The computed matrices at the end of one and two full turns are $-\mathbb{1}$ and $\mathbb{1}$. A full turn therefore multiplies ψ by -1, although the classical rotation has returned to the identity. This overall sign leaves the isolated state unchanged. It changes a relative phase if one coherent alternative undergoes the rotation while another does not: the combined amplitude changes from $\psi_1 + \psi_2$ to $-\psi_1 + \psi_2$.

Self-check. Consider rotation through $6\pi t^2$ about $n = (2, -1, 2)/3$ for $t \in I$. Find its lift from $(1, 0)$, compute its matrix at $t = 1/2$, and determine whether the rotation loop contracts.

Answer. The axis has norm one. The half-angle formula gives

$$(a(t), v(t)) = \big(\cos(3\pi t^2), \sin(3\pi t^2) n\big), \qquad U(1/2) = \frac{\sqrt{2}}{6} \begin{pmatrix} -3 - 2\mathrm{i} & 1 - 2\mathrm{i} \\ -1 - 2\mathrm{i} & -3 + 2\mathrm{i} \end{pmatrix}.$$

At $t = 1$ the lifted point is $(-1, 0)$, since the half-angle is 3π. The projected rotation is the identity, but its lift does not close. The closed-lift test therefore excludes a contraction.

Self-check. State in one sentence what a covering map is and why we introduced it to study loop classes. Explain why specifying one starting point upstairs is essential.

Answer. A covering map is a continuous surjection with disjoint local inverse sheets, introduced to lift paths and homotopies and to recover loop classes from their endpoints when the covering space is simply connected. The initial point selects a sheet on the first subinterval. Each successive endpoint selects the next sheet, which is the decisive step in path-lift existence and uniqueness.

Unit 4B Optional extensions

4.7 General map lifting and classification *(optional)*

The later computations of fundamental groups by open pieces do not depend on the general map-lifting criterion or the classification theorem in this section.

Definition 4.12 (Local path-connectedness)**.** A space is *locally path-connected* if every neighbourhood of each point contains a path-connected open neighbourhood of that point.

Let $p(z) = z^3$ and $f(z) = z^6$ on S^1, with base point 1. The induced maps multiply winding by 3 and 6 respectively. A lift is $\tilde{f}(z) = z^2$, since $(z^2)^3 = z^6$ and $\tilde{f}(1) = 1$. In contrast, $z \mapsto z^5$ has no such lift. Dividing the angle by three along the generating loop ends at $e^{2\pi \mathrm{i} 5/3} \neq 1$ and therefore does not define a loop upstairs. The closed-lift test rules out a global lift, although every small arc has inverse branches.

The following *lifting criterion* is stated with its proof sketched. For path-connected, locally path-connected Y, a continuous map $f\colon (Y, y_0) \to (X, x_0)$ lifts with value $\tilde{x}_0$ at y_0 exactly when

$$f_*\pi_1(Y, y_0) \subseteq p_*\pi_1(\widetilde{X}, \tilde{x}_0).$$

Necessity follows from $f_* = p_*\tilde{f}_*$. For sufficiency, choose a path from y_0 to y and lift its image under f. Define $\tilde{f}(y)$ as the lifted endpoint. The inclusion and the closed-lift test make this independent of the chosen path. On a small path-connected open set mapping into an evenly covered set, this rule is one inverse branch composed with f, so is continuous. This is Hatcher, Proposition 1.33.

For these power maps, the criterion reads $6\mathbb{Z} \subseteq 3\mathbb{Z}$ in the first case. In the second case, $5\mathbb{Z}$ is not contained in $3\mathbb{Z}$.

Definition 4.13 (Universal cover)**.** A simply connected covering space is called a *universal cover*.

The circle projection $\mathbb{R} \to S^1$, the antipodal covers $S^n \to \mathbb{R}\mathrm{P}^n$ for $n \geq 2$, and the rotation cover $S^3 \to \mathrm{SO}(3)$ are examples.

Definition 4.14 (Semilocal simple connectedness)**.** A space is *semilocally simply connected* if each $x \in X$ has an open neighbourhood U such that every loop in U based at x contracts in X.

The neighbourhood itself need not be simply connected.

Definition 4.15 (Covering isomorphism)**.** A *covering isomorphism* is a homeomorphism between the total spaces commuting with their projections to X. In the based version it also preserves the chosen lifted point.

Definition 4.16 (Conjugate subgroups)**.** Two subgroups are *conjugate* if they have the form H and $k^{-1}Hk$ for some group element k.

Theorem 4.17 (Existence and classification · *cited*)**.** *Let X be nonempty, connected, locally path-connected and semilocally simply connected. It has a universal cover. Isomorphism classes of based connected coverings correspond to subgroups of $\pi_1(X, x_0)$ by $p_*\pi_1(\widetilde{X}, \tilde{x}_0)$. Without chosen base points, they correspond to conjugacy classes of subgroups.*

This is Hatcher, Theorem 1.38. For the circle, the covers $z \mapsto z^m$ with $m \geq 1$ give the subgroups $m\mathbb{Z}$, whereas $\mathbb{R} \to S^1$ gives the zero subgroup. Local path-connectedness makes path components open, so connectedness implies path-connectedness here, both downstairs and in a connected cover.

Self-check. Classify the connected coverings of the circle up to covering isomorphism.

Answer. The circle satisfies the classification theorem's hypotheses: small open arcs are intervals. Every subgroup of $\mathbb{Z}$ is $m\mathbb{Z}$ for exactly one integer $m \geq 0$. For a nonzero subgroup H, let m be its least positive element. For every $h \in H$, integer division writes $h = qm + r$ with $0 \leq r < m$. Closure gives $r = h - qm \in H$, so minimality forces $r = 0$. Thus $H = m\mathbb{Z}$. The zero subgroup is $0\mathbb{Z}$. For $m \geq 1$, the map $z \mapsto z^m$ on S^1 induces multiplication by m on winding, so its image subgroup is $m\mathbb{Z}$. The cover $\mathbb{R} \to S^1$ has subgroup zero because $\mathbb{R}$ is contractible. The cited classification proves these exhaust the connected coverings. Conjugation in additive notation gives $-k + H + k = H$ for every subgroup $H \subseteq \mathbb{Z}$ and every $k \in \mathbb{Z}$, because addition commutes. Thus distinct subgroups give distinct unbased covering

isomorphism classes.

Summary. Local inverse sheets lift paths and their homotopies uniquely. For a simply connected covering space, based loop classes correspond to fibre points. A transitive deck group gives the same group structure after choosing a lifted base point. The antipodal cover computes $\pi_1(\mathbb{R}\mathrm{P}^n) \cong \mathbb{Z}_2$ for $n \geq 2$, using the sketched sphere result. In $\mathrm{SO}(3)$, a full turn is nontrivial and a double turn is trivial. Lecture 5 computes fundamental groups by joining open pieces.

Reading. Hatcher, *Algebraic Topology*, Proposition 1.14, §1.3, especially Propositions 1.30–1.34 and Theorem 1.38; Munkres, *Topology*, §§53–54. Nakahara, *Geometry, Topology and Physics*, §4.6.5 and Table 4.1.